Selected files

7 printable files

```
week0/task1.py
week0/task2.py
week0/task3.py
week0/task4.py
week0/task5.py
week0/task6.py
week0/task7.py
```

week0/task1.py

```python
import pandas as pd

df = pd.read_csv("airq402.csv")

print(df.head())

encoded = pd.get_dummies(df, columns=["City1", "City2"], dtype=int)

print(encoded.head())
```

week0/task2.py

```python
import pandas as pd

df = pd.read_csv("airq402.csv")

print(len(df))

df.dropna(axis="index", inplace=True)

print(len(df))
```

week0/task3.py

```python
import pandas as pd

df = pd.read_csv("airq402.csv")

train = df.sample(frac=0.8)
test = df.drop(train.index)

print(train.head(), len(train))
print(test.head(), len(test))
```

week0/task4.py

```python
import numpy

matrix = numpy.random.rand(5, 5)

print(f"Random matrix: \n{matrix}")

replaced_matrix = numpy.where(matrix > 0.5, 1, matrix)
```

```
8
9  print(f"New matrix: \n{replaced_matrix}")
```

week0/task5.py

```
1  # Load a CSV file using Pandas, display its summary statistics, and drop columns that have more than 50% missing values.
2
3  import pandas as pd
4
5  df = pd.read_csv("winequality-white.csv")
6  print(df)
7
8  print(df.describe(include="all"))
9
10 df = df.dropna(thresh=len(df) * 0.5, axis="columns")
11
12 print(df)
```

week0/task6.py

```
1  #6. Create a 2x2 grid of plots showing the distribution of four different features in your dataset.
2  import pandas as pd
3  import matplotlib.pyplot as plt
4
5  df = pd.read_csv("winequality-white.csv",sep=';')
6
7  df.hist(column=["fixed acidity", "volatile acidity", "citric acid", "residual sugar"], layout=(2, 2))
8  plt.show()
```

week0/task7.py

```
1  import pandas as pd
2
3  df = pd.read_csv("winequality-red.csv", sep=";")
4
5  filtered_df = df.dropna(axis="index")
6  flitered_df = df.drop_duplicates
7
8  features = filtered_df[["fixed acidity","volatile acidity", "citric acid", "residual sugar", "chlorides", "free sulfur dioxide",
   "total sulfur dioxide", "density", "pH", "sulphates", "alcohol"]]
9  target = filtered_df["quality"]
10
11 print(features)
12 print(target)
```